

An Exact Decomposition of the Divisor Summatory Function and Fixed- j Asymptotics

GUEHERN HWANG

Department of Mathematics, Ajou University

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ABSTRACT. In this paper, we study a decomposition of the divisor summatory function $D(x)$. We establish the exact identity

$$D(x) = 1 + \sum_{i \geq 1} (i+1)\pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1)T_j(x),$$

which decomposes $D(x)$ via inclusion–exclusion according to the number of distinct prime factors of n . The three terms correspond respectively to $n = 1$, prime powers, and integers with $\omega(n) \geq 2$.

We then establish an asymptotic formula with an explicit error term for $T_j(x)$ with fixed $j \geq 2$. We consider the two-variable Dirichlet series $F(s, y)$ and obtain the factorization $F(s, y) = \zeta(s)^{2+2y} H(s, y)$, where H is holomorphic near $s = 1$. Applying the Selberg–Delange method—a truncated Perron formula, contour deformation, and local Hankel analysis—we prove that

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log \log x)^j \quad \text{as } x \rightarrow \infty.$$

The fixed- j estimates do not by themselves yield an asymptotic formula for $D(x)$; recovering $D(x)$ from the identity would require estimates uniform in j , or a direct analysis of the generating function near $y = -1$.

1 Introduction

1.1 The Dirichlet divisor problem

Let $d(n)$ denote the number of divisors of a positive integer n , and let

$$D(x) := \sum_{n \leq x} d(n) \tag{1.1}$$

be its summatory function. Dirichlet [2] proved in 1849 that

$$D(x) = x \log x + (2\gamma - 1)x + \Delta(x), \tag{1.2}$$

where γ is the Euler–Mascheroni constant and $\Delta(x) = O(x^{1/2})$. The Dirichlet divisor problem asks for the infimum θ of all real numbers such that $\Delta(x) = O(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$.

Hardy [3] proved that $\theta \geq 1/4$, and it is conjectured that $\theta = 1/4$. In the other direction, Voronoï [5] proved the classical estimate

$$\Delta(x) = O(x^{1/3} \log x), \quad (1.3)$$

which implies $\theta \leq 1/3$. Further refinements eventually led to the classical published bound of Huxley [4]:

$$\theta \leq 131/416. \quad (1.4)$$

In this paper, rather than pursuing further improvements of θ , we explore an alternative approach: an exact combinatorial decomposition of $D(x)$ via inclusion–exclusion on the number of distinct prime factors of n .

1.2 A new decomposition

For a positive integer n , let $\omega(n)$ denote the number of distinct prime factors of n . For each integer $j \geq 2$, define

$$T_j(x) := \sum_{n \leq x} d(n) \binom{\omega(n)}{j} \quad (1.5)$$

where $\binom{\omega(n)}{j}$ is the binomial coefficient. The main identity of this paper is the following.

Theorem 1.1. For every real number $x \geq 1$,

$$D(x) = 1 + \sum_{i \geq 1} (i+1) \pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x) \quad (1.6)$$

where $\pi(x)$ denotes the prime-counting function. Both sums are effectively finite for each fixed x : we have $\pi(x^{1/i}) = 0$ for all sufficiently large i , and $T_j(x) = 0$ once j exceeds the maximum possible value of $\omega(n)$ among integers $n \leq x$.

The three terms on the right-hand side of (1.6) correspond, respectively, to $n = 1$, prime powers $\omega(n) = 1$, and integers with $\omega(n) \geq 2$. The proof of Theorem 1.1 is given in Section 2.

Classical results of Dirichlet give the asymptotic $D(x) \sim x \log x$. By the prime number theorem [1], the prime-power contribution satisfies

$$\sum_{i \geq 1} (i+1) \pi(x^{1/i}) \sim \frac{2x}{\log x}, \quad (1.7)$$

which is of strictly smaller order than $x \log x$. Hence the first two terms in the identity are negligible compared with the main asymptotic of $D(x)$, and the third sum

$$\sum_{j \geq 2} (-1)^j (j-1) T_j(x) \quad (1.8)$$

is responsible for the remaining main scale after cancellation. Individually, each $T_j(x)$ grows rapidly with j , but the alternating coefficients $(-1)^j (j-1)$ induce substantial cancellation in the full finite sum. The fixed- j asymptotics proved below give information about individual terms, but they do not by themselves determine this whole alternating sum.

Example 1.2. At $x = 10000$, direct computation gives

$$\begin{aligned} T_2(10000) &= 297,078 & T_3(10000) &= 140,640 \\ T_4(10000) &= 26,674 & T_5(10000) &= 1,216 \end{aligned}$$

and $T_j(10000) = 0$ for $j \geq 6$. The prime-power contribution equals 2,711, and the identity (1.6) yields

$$1 + 2,711 + (T_2 - 2T_3 + 3T_4 - 4T_5) = 1 + 2,711 + 90,956 = 93,668 = D(10000),$$

confirming Theorem 1.1 numerically. Moreover, the ratio between $T_j(x)$ and the leading term later proved in Theorem 1.3, $\frac{2^j}{j!} x \log x (\log \log x)^j$, converges to 1 only very slowly: at $x = 10^9$, this ratio is 0.43 for $j = 2$ and 0.21 for $j = 3$. This slow convergence motivates the refined asymptotic expansion of $T_j(x)$ developed in Section 6.

1.3 Main results

To analyze $T_j(x)$, we consider the two-variable Dirichlet series

$$F(s, y) := \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}, \quad (1.9)$$

which encodes $T_j(x)$ as the coefficient of y^j in the generating relation $\sum_{j \geq 0} T_j(x) y^j$. Here, we extend the definition of $T_j(x)$ to all $j \geq 0$ by the formula (1.5), so that $T_0(x) = D(x)$ and $T_1(x) = \sum_{n \leq x} d(n) \omega(n)$. The series $F(s, y)$ admits the factorization

$$F(s, y) = \zeta(s)^{2+2y} H(s, y), \quad (1.10)$$

where $H(s, y)$ is holomorphic in s on a neighborhood of $\operatorname{Re}(s) \geq 1$ and uniformly bounded for y in a small disk around 0. This factorization isolates the singularity of $F(s, y)$ at $s = 1$, and it is the starting point of the Selberg–Delange method [1].

Our main asymptotic result is the following.

Theorem 1.3. For each fixed integer $j \geq 2$, as $x \rightarrow \infty$,

$$\begin{aligned} T_j(x) &= \frac{2^j}{j!} x \log x (\log \log x)^j \\ &\quad + \frac{2^{j-1}}{(j-1)!} (\partial_y H(1, 0) - 2(1-\gamma)) x \log x (\log \log x)^{j-1} \\ &\quad + O(x \log x (\log \log x)^{j-2}). \end{aligned} \quad (1.11)$$

In particular,

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log \log x)^j. \quad (1.12)$$

The proof of Theorem 1.3 is given in Sections 4–6. The essential ingredients are the truncated Perron formula, contour deformation around $s = 1$, and a local Hankel analysis in the w -plane obtained via the substitution $w = (s-1) \log x$. The coefficient $\partial_y H(1, 0)$ admits an explicit expression as an absolutely convergent series over primes; we record this expression in Section 6, where it is used in the proof of Theorem 1.3.

Theorem 1.3 is a fixed- j result. Combining it with the identity (1.6) does not by itself give an asymptotic formula for $D(x)$, because the identity involves values of j that grow with x . In terms of the generating function $A(x, y) = \sum_{j \geq 0} T_j(x) y^j$, the alternating combination in the identity is governed by values near $y = -1$, whereas the analytic work below is carried out in a small disk around $y = 0$. Recovering $D(x)$ from the identity is therefore left as a future problem, requiring estimates uniform in j or a direct analysis near $y = -1$.

1.4 Outline of the paper

The paper is organized as follows. In Section 2, we prove the identity (Theorem 1.1) by partitioning $D(x)$ according to $\omega(n)$ and applying an inclusion–exclusion argument to the composite part. In Section 3, we introduce the two-variable Dirichlet series $F(s, y)$ and establish its factorization $F(s, y) = \zeta(s)^{2+2y} H(s, y)$, together with the analytic properties of $H(s, y)$ needed in the sequel.

Sections 4–6 are devoted to the proof of Theorem 1.3. In Section 4, we apply the truncated Perron formula to the generating partial sum $A(x, y) := \sum_{n \leq x} d(n)(1 + y)^{\omega(n)}$ and deform the contour around $s = 1$. In Section 5, we carry out the local Hankel analysis via the substitution $w = (s - 1) \log x$ and evaluate the main and lower-order contributions. In Section 6, we extract the asymptotic formula for $T_j(x)$ from that of $A(x, y)$ by differentiating in y , thereby completing the proof of Theorem 1.3.

Finally, in Section 7, we discuss why the fixed- j asymptotics and the identity do not yet determine the full alternating sum for $D(x)$, and what additional uniform estimates would be needed.

2 Proof of the identity

In this section, we prove the identity stated in Theorem 1.1. The proof proceeds by partitioning the sum $D(x) = \sum_{n \leq x} d(n)$ according to the value of $\omega(n)$, and then applying an inclusion–exclusion argument to the composite part.

Proof of Theorem 1.1. For fixed x , all sums below are finite. Indeed, $\pi(x^{1/i}) = 0$ for all sufficiently large i , and $\binom{\omega(n)}{j} = 0$ whenever $j > \omega(n)$.

Partition the integers n with $1 \leq n \leq x$ into three disjoint classes:

- (i) $n = 1$ (equivalently, $\omega(n) = 0$);
- (ii) prime powers, i.e., $\omega(n) = 1$;
- (iii) composite integers with $\omega(n) \geq 2$.

Writing $D(x)$ as the sum of contributions from each class, we have

$$D(x) = d(1) + \sum_{\substack{n \leq x \\ \omega(n)=1}} d(n) + \sum_{\substack{n \leq x \\ \omega(n) \geq 2}} d(n). \quad (2.1)$$

Since $d(1) = 1$, the first term equals 1.

For the second term, every integer n with $\omega(n) = 1$ is of the form $n = p^i$ for a unique prime p and a unique integer $i \geq 1$, and $d(p^i) = i + 1$. The number of prime powers $p^i \leq x$ with exponent exactly i equals $\pi(x^{1/i})$. Hence

$$\sum_{\substack{n \leq x \\ \omega(n)=1}} d(n) = \sum_{i \geq 1} (i + 1) \pi(x^{1/i}). \quad (2.2)$$

For the third term, we apply inclusion–exclusion on the set of distinct prime divisors of n . For each integer $j \geq 2$, recall the definition

$$T_j(x) = \sum_{n \leq x} d(n) \binom{\omega(n)}{j}.$$

We claim that

$$\sum_{\substack{n \leq x \\ \omega(n) \geq 2}} d(n) = \sum_{j \geq 2} (-1)^j (j-1) T_j(x). \quad (2.3)$$

To prove (2.3), interchange the order of summation on the right-hand side. This involves only finite sums, so no convergence issue occurs:

$$\sum_{j \geq 2} (-1)^j (j-1) T_j(x) = \sum_{n \leq x} d(n) \sum_{j \geq 2} (-1)^j (j-1) \binom{\omega(n)}{j}.$$

For n with $\omega(n) \in \{0, 1\}$, we have $\binom{\omega(n)}{j} = 0$ for all $j \geq 2$, so such n do not contribute. For n with $\omega(n) = m \geq 2$, we use the following combinatorial identity.

Lemma 2.1. For every integer $m \geq 2$,

$$\sum_{j=2}^m (-1)^j (j-1) \binom{m}{j} = 1. \quad (2.4)$$

Proof of the lemma. Starting from the standard identities

$$\sum_{j=0}^m (-1)^j \binom{m}{j} = 0 \quad \text{and} \quad \sum_{j=0}^m (-1)^j j \binom{m}{j} = \frac{d}{dx} (1-x)^m \Big|_{x=1} = 0 \quad (\text{for } m \geq 2),$$

we combine them to obtain

$$\sum_{j=0}^m (-1)^j (j-1) \binom{m}{j} = 0 \quad (\text{for } m \geq 2).$$

Separating the $j = 0$ and $j = 1$ terms:

$$1 \cdot (-1) + 0 \cdot \binom{m}{1} + \sum_{j=2}^m (-1)^j (j-1) \binom{m}{j} = 0,$$

which yields

$$\sum_{j=2}^m (-1)^j (j-1) \binom{m}{j} = 1. \quad \square$$

Applying the lemma, we obtain

$$\sum_{n \leq x} d(n) \sum_{j \geq 2} (-1)^j (j-1) \binom{\omega(n)}{j} = \sum_{\substack{n \leq x \\ \omega(n) \geq 2}} d(n) \cdot 1,$$

which establishes (2.3).

Combining the contributions from all three classes yields

$$D(x) = 1 + \sum_{i \geq 1} (i+1) \pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x),$$

completing the proof. $\square$

3 Dirichlet series and factorization

In this section, we introduce the two-variable Dirichlet series $F(s, y)$ and establish its key analytic properties. The main output is the factorization $F(s, y) = \zeta(s)^{2+2y} H(s, y)$, where $H(s, y)$ is holomorphic in a neighborhood of $s = 1$ and uniformly bounded in a small disk around $y = 0$.

3.1 Euler product

Recall the definition

$$F(s, y) := \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}. \quad (3.1)$$

Since the arithmetic function $n \mapsto d(n)(1+y)^{\omega(n)}$ is multiplicative in n for each fixed y , the series admits an Euler product representation for $\operatorname{Re}(s) > 1$ [1].

Lemma 3.1. For $\operatorname{Re}(s) > 1$ and every $y \in \mathbb{C}$,

$$F(s, y) = \prod_p \left(1 + (1+y) \left((1-p^{-s})^{-2} - 1 \right) \right), \quad (3.2)$$

where the product is over all primes p .

Proof. By multiplicativity, for $\operatorname{Re}(s) > 1$ we have

$$F(s, y) = \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s} = \prod_p \left(\sum_{k \geq 0} \frac{d(p^k)(1+y)^{\omega(p^k)}}{p^{ks}} \right).$$

For $k = 0$, the summand equals 1. For $k \geq 1$, we have $d(p^k) = k + 1$ and $\omega(p^k) = 1$, so the local factor becomes

$$1 + (1+y) \sum_{k \geq 1} \frac{k+1}{p^{ks}}.$$

Setting $u := p^{-s}$ and using the standard identity

$$\sum_{k \geq 1} (k+1)u^k = \frac{1}{(1-u)^2} - 1 \quad \text{for } |u| < 1,$$

the local factor equals $1 + (1+y)((1-p^{-s})^{-2} - 1)$, which proves (3.2). $\square$

3.2 Factorization isolating the singularity

The Riemann zeta function admits the Euler product [1] $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ for $\operatorname{Re}(s) > 1$, so that $\zeta(s)^{2+2y} = \prod_p (1 - p^{-s})^{-(2+2y)}$ on the same half-plane, provided a branch of the logarithm is fixed. We use the principal branch, which is analytic in s in a neighborhood of any point with $\operatorname{Re}(s) > 1$ and in y in a neighborhood of 0.

Define

$$H(s, y) := \prod_p (1 - p^{-s})^{2+2y} \left(1 + (1+y) \left((1 - p^{-s})^{-2} - 1 \right) \right). \quad (3.3)$$

Theorem 3.2. For $\operatorname{Re}(s) > 1$ and y in a neighborhood of 0,

$$F(s, y) = \zeta(s)^{2+2y} H(s, y). \quad (3.4)$$

Proof. This follows by combining Lemma 3.1 with the Euler product for $\zeta(s)^{2+2y}$ and regrouping factors. $\square$

The purpose of the factorization (3.4) is to isolate the singular behavior of $F(s, y)$ at $s = 1$ into the factor $\zeta(s)^{2+2y}$, leaving $H(s, y)$ as a well-behaved function in a larger region, as we now establish.

3.3 Analytic properties of $H(s, y)$

We show that the Euler product (3.3) defines a holomorphic function on a strip containing $\operatorname{Re}(s) = 1$, uniformly for y in a small disk around 0. The key is that each local factor $H_p(s, y)$ deviates from 1 by a quantity of order $p^{-2\operatorname{Re}(s)}$.

Lemma 3.3. Fix $\eta \in (0, 1/2)$ and $\delta \in (0, 1/2)$. Write $u := p^{-s}$ and denote the local factor by

$$H_p(s, y) := (1 - u)^{2+2y} (1 + (1 + y) ((1 - u)^{-2} - 1)).$$

Then for $|y| \leq \eta$ and $\operatorname{Re}(s) \geq 1 - \delta$,

$$H_p(s, y) = 1 + O_{\eta, \delta}(u^2), \tag{3.5}$$

where the implied constant depends only on η and δ .

Proof. The condition $\operatorname{Re}(s) \geq 1 - \delta$ implies that, for every prime $p \geq 2$,

$$|u| = p^{-\operatorname{Re}(s)} \leq p^{-(1-\delta)} \leq 2^{-(1-\delta)} < 1,$$

since $\delta < 1/2$. In particular, $|u|$ is bounded away from 1, and the power series expansions

$$(1 - u)^{-2} - 1 = 2u + 3u^2 + 4u^3 + \cdots = 2u + O_\delta(u^2),$$

$$(1 - u)^{2+2y} = 1 - (2 + 2y)u + \binom{2+2y}{2}u^2 + \cdots = 1 - (2 + 2y)u + O_{\eta, \delta}(u^2),$$

converge absolutely, with remainders uniform for $|y| \leq \eta$ and $\operatorname{Re}(s) \geq 1 - \delta$. Multiplying the expansions:

$$H_p(s, y) = (1 - (2 + 2y)u + O_{\eta, \delta}(u^2))(1 + (1 + y)(2u + O_\delta(u^2))).$$

The coefficient of u^1 on the right is $-(2 + 2y) + 2(1 + y) = 0$. Hence $H_p(s, y) = 1 + O_{\eta, \delta}(u^2)$, proving (3.5). $\square$

Proposition 3.4. Fix $\eta \in (0, 1/2)$ and $\delta \in (0, 1/2)$, and let

$$\mathcal{R}_{\delta, \eta} := \{(s, y) \in \mathbb{C}^2 : \operatorname{Re}(s) \geq 1 - \delta, |y| \leq \eta\}. \tag{3.6}$$

Then the partial products of the Euler product (3.3) converge absolutely and uniformly on $\mathcal{R}_{\delta, \eta}$ to a function $H(s, y)$. For every y with $|y| \leq \eta$, the function $H(s, y)$ is holomorphic in s on the open strip $\operatorname{Re}(s) > 1 - \delta$. Moreover, $H(s, y)$ is jointly holomorphic in (s, y) on $\operatorname{int}(\mathcal{R}_{\delta, \eta})$, and H is uniformly bounded on $\mathcal{R}_{\delta, \eta}$.

Proof. By Lemma 3.3, for every $(s, y) \in \mathcal{R}_{\delta, \eta}$ and every prime p ,

$$|H_p(s, y) - 1| \leq C_{\eta, \delta} p^{-2(1-\delta)}, \quad (3.7)$$

where the constant $C_{\eta, \delta}$ depends only on η and δ . Since $\delta < 1/2$, we have $2(1 - \delta) > 1$, and the series $\sum_n n^{-2(1-\delta)}$ converges; in particular, $\sum_p p^{-2(1-\delta)}$ converges. By the Weierstrass M -test, the series

$$\sum_p |H_p(s, y) - 1|$$

converges uniformly on $\mathcal{R}_{\delta, \eta}$. Hence the infinite product criterion applies: if $\sum_n |u_n|$ converges uniformly, then the product $\prod_n (1 + u_n)$ converges absolutely and uniformly. Therefore the partial products $\prod_{p \leq P} H_p(s, y)$ converge absolutely and uniformly on $\mathcal{R}_{\delta, \eta}$ to a limit, which we denote $H(s, y)$.

To deduce holomorphy in s , fix $y_0 \in \mathbb{C}$ with $|y_0| \leq \eta$ and let $s_0 \in \mathbb{C}$ be any point with $\operatorname{Re}(s_0) > 1 - \delta$. Choose $r > 0$ small enough that $r < \operatorname{Re}(s_0) - (1 - \delta)$, and define

$$K := \overline{D}(s_0, r) = \{s \in \mathbb{C} : |s - s_0| \leq r\}. \quad (3.8)$$

The set K is closed and bounded in $\mathbb{C}$, so by the Heine–Borel theorem it is compact. By the choice of r , every $s \in K$ satisfies

$$\operatorname{Re}(s) \geq \operatorname{Re}(s_0) - r > 1 - \delta,$$

and hence $K \times \{y_0\} \subset \mathcal{R}_{\delta, \eta}$. The uniform convergence of the partial products on $\mathcal{R}_{\delta, \eta}$ established above therefore restricts to uniform convergence of $\prod_{p \leq P} H_p(s, y_0)$ on K as $P \rightarrow \infty$.

Each finite partial product $\prod_{p \leq P} H_p(s, y_0)$ is holomorphic in s on the open disk $D(s_0, r) = \operatorname{int}(K)$, since each factor $H_p(\cdot, y_0)$ is. By the theorem on uniform limits of holomorphic functions (Weierstrass), the limit $H(\cdot, y_0)$ is holomorphic on $D(s_0, r)$, and in particular at s_0 . Since s_0 with $\operatorname{Re}(s_0) > 1 - \delta$ was arbitrary, $H(\cdot, y_0)$ is holomorphic on the open strip $\operatorname{Re}(s) > 1 - \delta$. Since y_0 with $|y_0| \leq \eta$ was also arbitrary, the holomorphy claim in s is proved.

For joint holomorphy, each finite partial product $\prod_{p \leq P} H_p(s, y)$ is holomorphic in (s, y) on $\operatorname{int}(\mathcal{R}_{\delta, \eta})$, since each factor $H_p(s, y)$ is. The uniform convergence of the partial products on $\mathcal{R}_{\delta, \eta}$ established above restricts to uniform convergence on every compact subset of $\operatorname{int}(\mathcal{R}_{\delta, \eta})$, since $\operatorname{int}(\mathcal{R}_{\delta, \eta}) \subset \mathcal{R}_{\delta, \eta}$. By the theorem on uniform limits of holomorphic functions in two complex variables, the limit $H(s, y)$ is jointly holomorphic on $\operatorname{int}(\mathcal{R}_{\delta, \eta})$.

Uniform boundedness on $\mathcal{R}_{\delta, \eta}$ follows from (3.7):

$$|H(s, y)| \leq \prod_p (1 + |H_p(s, y) - 1|) \leq \exp \left(\sum_p |H_p(s, y) - 1| \right) \leq \exp \left(C_{\eta, \delta} \sum_p p^{-2(1-\delta)} \right),$$

the right-hand side being finite and independent of $(s, y) \in \mathcal{R}_{\delta, \eta}$. □

Lemma 3.5. The function $H(s, y)$ satisfies

$$H(s, 0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1/2. \quad (3.9)$$

In particular, $H(1, 0) = 1$.

Proof. On $\operatorname{Re}(s) > 1$, the factorization (3.4) specialized at $y = 0$ reads

$$F(s, 0) = \zeta(s)^2 H(s, 0).$$

On the other hand, the Dirichlet series defining F gives

$$F(s, 0) = \sum_{n \geq 1} \frac{d(n)}{n^s} = \zeta(s)^2 \quad \text{on } \operatorname{Re}(s) > 1.$$

Comparing the two expressions yields $\zeta(s)^2 H(s, 0) = \zeta(s)^2$ on $\operatorname{Re}(s) > 1$. Since $\zeta(s) \neq 0$ on $\operatorname{Re}(s) > 1$ (which follows directly from the Euler product representation of ζ), we may divide both sides by $\zeta(s)^2$ to obtain

$$H(s, 0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1.$$

By Proposition 3.4, applied with any $\delta \in (0, 1/2)$, the function $H(s, 0)$ is holomorphic on $\operatorname{Re}(s) > 1 - \delta$; letting $\delta \rightarrow 1/2$ shows that $H(s, 0)$ is holomorphic on the connected open set $\operatorname{Re}(s) > 1/2$. The constant function 1 is also holomorphic there, and the two functions agree on the open subset $\operatorname{Re}(s) > 1$. By the identity theorem, they agree on all of $\operatorname{Re}(s) > 1/2$. In particular, $H(1, 0) = 1$. $\square$

4 Perron formula and contour deformation

To extract the asymptotic behavior of $T_j(x)$ from the Dirichlet series $F(s, y)$, we work with the generating partial sum

$$A(x, y) := \sum_{n \leq x} d(n)(1 + y)^{\omega(n)}, \quad (4.1)$$

which satisfies the generating relation

$$A(x, y) = \sum_{j \geq 0} T_j(x) y^j. \quad (4.2)$$

Throughout this section, we initially fix one parameter:

- $\eta \in (0, 1/2)$, controlling the size of the disk $|y| \leq \eta$.

After the contour depth δ' is fixed, we may shrink η if necessary for the contour estimates.

The fixed strip depth δ' and the truncation-dependent quantity λ_T will be chosen together in Section 4.3. This keeps the Euler factor estimates, the zero-free region, and the contour growth bound tied to the same strip parameter.

We also abbreviate $\alpha := 2 + 2y$. Thus the factorization (3.4) becomes $F(s, y) = \zeta(s)^\alpha H(s, y)$. For $|y| \leq \eta$ with $\eta < 1/2$, the exponent α ranges in a disk of radius $2\eta < 1$ around 2, so in particular α is bounded away from non-positive integers.

4.1 The truncated Perron formula

We quote the following standard form of the truncated Perron formula (see, e.g., [1]).

Theorem 4.1 (Truncated Perron formula). Let $f(s) = \sum_{n \geq 1} a_n n^{-s}$ be a Dirichlet series with abscissa of absolute convergence σ_a . For real numbers $c > \sigma_a$, $x \geq 2$, and $T \geq 1$ with $x \notin \mathbb{Z}$,

$$\sum_{n \leq x} a_n = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} f(s) \frac{x^s}{s} ds + R(x, T), \quad (4.3)$$

where the truncation error satisfies

$$R(x, T) \ll x^c \sum_{n \geq 1} \frac{|a_n|}{n^c} \frac{1}{1 + T \left| \log \frac{x}{n} \right|}. \quad (4.4)$$

If $x = N$ is an integer, we do not apply the Perron formula directly at N . Instead, we apply it at

$$X := N + \frac{1}{2}.$$

Then $X \notin \mathbb{Z}$ and

$$A(N, y) = A(X, y),$$

because the integers satisfying $n \leq N$ are exactly the integers satisfying $n \leq N + 1/2$. Moreover,

$$X = N + O(1), \quad \log X = \log N + O(1/N),$$

so every asymptotic estimate obtained for X transfers back to N . Thus it is enough in the Perron argument to work with non-integral x ; the integer case follows from this replacement.

We apply Theorem 4.1 to the Dirichlet series $F(s, y) = \sum_{n \geq 1} d(n)(1+y)^{\omega(n)} n^{-s}$. For $|y| \leq \eta$ with η sufficiently small, the series converges absolutely for $\operatorname{Re}(s) > 1$, so we may take any $c > 1$. Choose, for instance, $c = 1 + 1/\log x$, which is the standard choice to balance the main term size and the truncation error.

For $x \geq 2$ with $x \notin \mathbb{Z}$, $|y| \leq \eta$, and $T \geq 1$, put $c = 1 + 1/\log x$. Theorem 4.1 gives

$$A(x, y) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} ds + E_{\text{Perron}}(x, T, y), \quad (4.5)$$

where

$$E_{\text{Perron}}(x, T, y) \ll_{\eta} x^c \sum_{n \geq 1} \frac{d(n)(1+\eta)^{\omega(n)}}{n^c} \frac{1}{1 + T \left| \log \frac{x}{n} \right|}. \quad (4.6)$$

Indeed, applying (4.4) with $a_n = d(n)(1+y)^{\omega(n)}$ and using $|1+y| \leq 1+\eta$ gives the displayed bound.

4.2 Analytic preliminaries

To control the contour integrals arising from the deformation in Section 4.3, we will need:

- (a) a region in which $\zeta(s)$ does not vanish, so that the contour can avoid its zeros;
- (b) a continuous branch of $\zeta(s)^{2+2y}$ on a neighborhood of the contour;
- (c) a uniform bound on $|\zeta(s)|$ along vertical lines in the critical strip;
- (d) a corresponding bound on $\arg \zeta(s)$ on the chosen branch.

We collect these analytic ingredients here, before constructing the contour itself in Section 4.3.

Zero-free region

For the contour deformation to be valid, we need a region in which $\zeta(s) \neq 0$. We use the following classical result of de la Vallée Poussin, quoted without proof; see, e.g., [1].

Theorem 4.2 (de la Vallée Poussin zero-free region). There exists an absolute constant $c_0 \in (0, 1/2)$ such that $\zeta(s) \neq 0$ on the open region

$$\mathcal{Z} := \left\{ s = \sigma + it \in \mathbb{C} : \sigma > 1 - \frac{c_0}{\log(|t| + 2)} \right\} \setminus \{1\}. \quad (4.7)$$

The boundary of $\mathcal{Z}$ is curved, and integrating along it would complicate the estimation of the resulting contour pieces. We will instead use a fixed vertical line lying strictly inside $\mathcal{Z}$ for all $|t| \leq T$; the precise depth of this line is fixed in Section 4.3.

Continuous branch of $\zeta(s)^\alpha$

To make sense of $\zeta(s)^\alpha = \zeta(s)^{2+2y}$ on a neighborhood of the contour, we specify a branch on the region

$$\mathcal{D} := \mathcal{Z} \setminus \mathcal{S}, \quad (4.8)$$

where $\mathcal{S} := \{s \in \mathbb{C} : \Im(s) = 0, \operatorname{Re}(s) \leq 1\}$ is the slit along the real axis to the left of $s = 1$. The slit $\mathcal{S}$ disconnects the segment $\mathcal{Z} \cap \mathcal{S}$ from the rest of $\mathcal{Z}$ from above and below, rendering $\mathcal{D}$ simply connected.¹

On $\mathcal{D}$, both $\zeta(s)$ and $\zeta'(s)$ are holomorphic, and $\zeta(s) \neq 0$. Fix any real number $c_* > 1$ (for example $c_* = 2$); since $\zeta(c_*) > 0$, the principal logarithm $\log \zeta(c_*) \in \mathbb{R}$ is well-defined. Define

$$L(s) := \log \zeta(c_*) + \int_{c_*}^s \frac{\zeta'(\omega)}{\zeta(\omega)} d\omega \quad (s \in \mathcal{D}), \quad (4.9)$$

where the integral is taken along any path in $\mathcal{D}$ from c_* to s . Since $\mathcal{D}$ is simply connected and ζ'/ζ is holomorphic on $\mathcal{D}$, the integral is path-independent, and L is a well-defined holomorphic function on $\mathcal{D}$.

We claim $e^{L(s)} = \zeta(s)$ on $\mathcal{D}$. Indeed, $L'(s) = \zeta'(s)/\zeta(s)$ on $\mathcal{D}$, so

$$\frac{d}{ds}(e^{-L(s)}\zeta(s)) = e^{-L(s)}(\zeta'(s) - L'(s)\zeta(s)) = 0.$$

Hence $e^{-L(s)}\zeta(s)$ is constant on $\mathcal{D}$, and evaluating at $s = c_*$ gives $e^{-L(c_*)}\zeta(c_*) = e^{-\log \zeta(c_*)}\zeta(c_*) = 1$. Therefore $e^{L(s)} = \zeta(s)$ throughout $\mathcal{D}$, so L serves as a continuous branch of $\log \zeta$ on $\mathcal{D}$.

We define

$$\zeta(s)^\alpha := e^{\alpha L(s)} \quad (s \in \mathcal{D}, \alpha \in \mathbb{C}). \quad (4.10)$$

This is single-valued and holomorphic in s on $\mathcal{D}$, and entire in α . At $\alpha = 2$, the definition reduces to $\zeta(s)^2$ in the usual sense.

Vertical growth of $\zeta(s)$

We use the following uniform bound on $|\zeta(s)|$, obtained by combining the convexity bound with the trivial estimate $\zeta(\sigma + it) \ll \log |t|$ for $\sigma \geq 1$ (see, e.g., [1] for the convexity bound).

Theorem 4.3 (Uniform bound on $\zeta(s)$ in a vertical strip). Fix $\delta \in (0, 1/2)$. For every $\varepsilon > 0$,

$$|\zeta(\sigma + it)| \ll_{\delta, \varepsilon} |t|^{\delta/2 + \varepsilon} \quad \text{for } 1 - \delta \leq \sigma \leq 1 + \delta \text{ and } |t| \geq 2, \quad (4.11)$$

where the implied constant is independent of t .

Sketch. For $1 - \delta \leq \sigma \leq 1$, the convexity bound gives $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{(1-\sigma)/2 + \varepsilon}$. Taking the supremum of $(1 - \sigma)/2$ over $\sigma \in [1 - \delta, 1]$ yields $\delta/2$, so $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{\delta/2 + \varepsilon}$ on this subrange.

For $1 \leq \sigma \leq 1 + \delta$, the standard estimate $|\zeta(\sigma + it)| \ll \log |t|$ holds, and $\log |t| \ll_\varepsilon |t|^\varepsilon \leq |t|^{\delta/2 + \varepsilon}$. Combining the two ranges gives (4.11). $\square$

¹Removing the slit from $s = 1$ to the boundary cuts the only possible loop around the puncture $s = 1$, just as $\mathbb{C} \setminus (-\infty, 0]$ is the standard simply connected branch domain for the logarithm. This is the topological input needed for path independence in the definition of $L(s)$.

Bound on $\arg \zeta$

We also use the following standard estimate on the branch determined by (4.9); see, e.g., [1].

Theorem 4.4 (Bound on $\arg \zeta$). There exists an absolute constant $C_{\arg} > 0$ such that, for every $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$,

$$|\arg \zeta(s)| \leq C_{\arg} \log |t|, \quad (4.12)$$

where the branch of $\arg \zeta$ is the one determined by the logarithm L on $\mathcal{D}$.

4.3 Contour decomposition

We now introduce the parameters governing the contour deformation and construct the deformed contour.

Depth of the deformation

Fix once and for all an absolute constant

$$\delta' \in \left(\frac{c_0}{2 \log 3}, \frac{c_0}{2} \right), \quad (4.13)$$

where c_0 is the constant of Theorem 4.2; the interval is non-empty since $\log 3 > 1$. This range of δ' is chosen so that the deformed contour remains in the fixed strip used in Section 4.4.

For each truncation height $T \geq 1$, define

$$\lambda_T := \frac{c_0}{2 \log(T+2)}. \quad (4.14)$$

By construction, $\lambda_T \rightarrow 0$ as $T \rightarrow \infty$, and for every $T \geq 1$,

$$\lambda_T \leq \frac{c_0}{2 \log 3} < \delta', \quad (4.15)$$

so the depth $\sigma_0(T) := 1 - \lambda_T$ of the deformed contour satisfies $\sigma_0(T) > 1 - \delta'$. Consequently, the contour $\mathcal{C}(T, \rho)$ defined below lies in $\mathcal{R}_{\delta', \eta}$ for every $T \geq 1$ and every $|y| \leq \eta$. In particular, Proposition 3.4 applies on the contour with $\delta = \delta'$.

Finally, the choice $c = 1 + 1/\log x$ in the Perron representation (4.5) satisfies $c \leq 1 + \delta'$ for all x sufficiently large, ensuring that the abscissa c lies inside the strip $1 - \delta' \leq \sigma \leq 1 + \delta'$ used in the contour estimates. We assume this hereafter.

The deformed contour

Fix a truncation height $T \geq 1$, and set

$$\sigma_0 := \sigma_0(T) := 1 - \lambda_T. \quad (4.16)$$

Choose a parameter $\rho > 0$ with $\rho < \lambda_T$; its precise choice as a function of x will be made in Section 5. The notation $i0^\pm$ below denotes boundary values: the two Hankel arms are first taken at heights $\pm \varepsilon$ with $\varepsilon > 0$, and the limit $\varepsilon \rightarrow 0^+$ is taken after the corresponding integrals are formed. Thus the approximating contours lie in the slit domain $\mathcal{D}$ before the boundary value limit is taken.

We deform the original Perron segment $[c - iT, c + iT]$ leftward, around the singularity at $s = 1$, into a contour $\mathcal{C} = \mathcal{C}(T, \rho)$ traversed from $c - iT$ to $c + iT$. It consists of the following five pieces:

- (1) $\mathcal{C}_{\text{bot}}$: the horizontal segment from $c - iT$ to $\sigma_0 - iT$.
- (2) $\mathcal{C}_{\text{left}}^-$: the vertical segment from $\sigma_0 - iT$ to $\sigma_0 + i0^-$.
- (3) $\mathcal{H}_{\text{loc}}$: a local Hankel contour around $s = 1$, consisting of
 - (3a) the lower arm from $\sigma_0 + i0^-$ to $1 - \rho + i0^-$;
 - (3b) the circle $|s - 1| = \rho$, traversed from $1 - \rho + i0^-$ to $1 - \rho + i0^+$ counterclockwise through the right side of $s = 1$;
 - (3c) the upper arm from $1 - \rho + i0^+$ to $\sigma_0 + i0^+$.
- (4) $\mathcal{C}_{\text{left}}^+$: the vertical segment from $\sigma_0 + i0^+$ to $\sigma_0 + iT$.
- (5) $\mathcal{C}_{\text{top}}$: the horizontal segment from $\sigma_0 + iT$ to $c + iT$.

The five pieces are joined consecutively, so $\mathcal{C}$ has the same initial and terminal points as the original Perron contour. The Hankel arms are understood as boundary values along the slit $\mathcal{S}$; in the local representation near $s = 1$, their discontinuity is carried by the factor $(s - 1)^{-2-2y}$ and produces the local contribution I_H .

Verification that the contour avoids ζ -zeros

We claim that $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$, where $\mathcal{Z}$ is the zero-free region of Theorem 4.2. For the horizontal pieces $\mathcal{C}_{\text{bot}}$ and $\mathcal{C}_{\text{top}}$,

$$\operatorname{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2 \log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \quad (4.17)$$

For the vertical pieces $\mathcal{C}_{\text{left}}^\pm$,

$$\operatorname{Re}(s) = \sigma_0(T) = 1 - \frac{c_0}{2 \log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \quad (4.18)$$

For the local Hankel contour $\mathcal{H}_{\text{loc}}$, the two horizontal arms (3a) and (3c) satisfy $\operatorname{Re}(s) \geq \sigma_0(T)$ and $|\Im s| \leq \rho$, hence are covered by the same estimate as the vertical pieces. For the small circle (3b),

$$\operatorname{Re}(s) \geq 1 - \rho > 1 - \lambda_T = 1 - \frac{c_0}{2 \log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \quad (4.19)$$

Therefore $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$. Combined with the boundary-value convention for the Hankel arms, the approximating contours with heights $\pm \varepsilon$ are contained in $\mathcal{D}$ away from $s = 1$. The integrand $F(s, y)x^s/s$ is therefore well-defined and holomorphic on a neighborhood of each approximating contour, and the contour $\mathcal{C}$ is understood as the corresponding boundary-value limit.

Cauchy's theorem

For $\varepsilon > 0$, replace the two Hankel arms along the slit by parallel paths at heights $-\varepsilon$ and $+\varepsilon$, and denote the resulting deformed contour by $\mathcal{C}_\varepsilon$. The Perron contour and $\mathcal{C}_\varepsilon$ share the same endpoints $c \pm iT$. Together, the Perron contour traversed from $c - iT$ to $c + iT$ and $\mathcal{C}_\varepsilon$ traversed in the reverse direction form a closed loop Γ_ε , oriented counterclockwise. The keyhole formed by the approximating Hankel contour and the slit $\mathcal{S}$ excludes the point $s = 1$ from the interior of Γ_ε , so the interior of Γ_ε is contained in the rectangle

$$\mathcal{R}_\Gamma := \{s = \sigma + it : \sigma_0(T) \leq \sigma \leq c, |t| \leq T\} \setminus \{1\}, \quad (4.20)$$

with the slit further excluded. For every $s \in \mathcal{R}_\Gamma$,

$$\operatorname{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2 \log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}, \quad (4.21)$$

so $\mathcal{R}_\Gamma \subset \mathcal{Z}$. Combined with the on-contour verifications (4.17)–(4.19), this shows that Γ_ε together with its interior, with the slit removed, is contained in the slit domain $\mathcal{D}$. Hence $F(s, y)x^s/s$ is holomorphic on an open neighborhood of Γ_ε and its interior. By Cauchy's theorem,

$$\oint_{\Gamma_\varepsilon} F(s, y) \frac{x^s}{s} ds = 0, \quad (4.22)$$

which yields

$$\int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} ds = \int_{\mathcal{C}_\varepsilon} F(s, y) \frac{x^s}{s} ds.$$

Letting $\varepsilon \rightarrow 0^+$ gives the boundary-value contour $\mathcal{C}$ and hence

$$\int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} ds = \int_{\mathcal{C}} F(s, y) \frac{x^s}{s} ds. \quad (4.23)$$

Decomposition of $A(x, y)$

For non-integral x , combining (4.23) with the Perron representation (4.5), we obtain

$$A(x, y) = I_H + I_{\text{top}} + I_{\text{bot}} + I_{\text{left}} + E_{\text{Perron}}(x, T, y), \quad (4.24)$$

where, for $* \in \{\text{top}, \text{bot}, \text{left}\}$,

$$I_* := \frac{1}{2\pi i} \int_{\mathcal{C}_*} F(s, y) \frac{x^s}{s} ds, \quad (4.25)$$

with $\mathcal{C}_{\text{left}} := \mathcal{C}_{\text{left}}^- \cup \mathcal{C}_{\text{left}}^+$, and

$$I_H := \frac{1}{2\pi i} \int_{\mathcal{H}_{\text{loc}}} F(s, y) \frac{x^s}{s} ds. \quad (4.26)$$

Writing $\mathcal{H}_{\text{loc}} = \mathcal{H}_{\text{loc}}^- \cup \mathcal{H}_{\text{loc}}^\circ \cup \mathcal{H}_{\text{loc}}^+$ for the lower arm, the small circle, and the upper arm, respectively, fixes the geometric decomposition of the local contour. The lower arm is oriented from $\sigma_0 + i0^-$ to $1 - \rho + i0^-$, the small circle is oriented counterclockwise from $1 - \rho + i0^-$ to $1 - \rho + i0^+$ through the right side of $s = 1$, and the upper arm is oriented from $1 - \rho + i0^+$ to $\sigma_0 + i0^+$.

The local Hankel contribution I_H contains the main asymptotic behavior and will be evaluated in Section 5. The remaining contributions $I_{\text{top}}, I_{\text{bot}}, I_{\text{left}}$ are estimated in Section 4.4 and shown to be of lower order than the main term, after a suitable choice of $T = T(x)$.

4.4 Estimates of the contour pieces

In this section, we estimate the three nonlocal contour contributions $I_{\text{top}}, I_{\text{bot}}, I_{\text{left}}$ appearing in the decomposition (4.24), and combine them with the Perron truncation error E_{Perron} to obtain a single error bound. The local Hankel contribution I_H is deferred to Section 5.

We first collect the two bounds for $F(s, y)$ used below.

Lemma 4.5 (Bounds for $F(s, y)$ on the deformed contour). Let $0 < \eta < 1/2$, and let δ' be fixed as in (4.13). Let $\varepsilon > 0$ and put $\alpha = 2 + 2y$. Define

$$B := B(\eta, \delta', \varepsilon) := (2 + 2\eta) \left(\frac{\delta'}{2} + \varepsilon \right) + 2\eta C_{\text{arg}}. \quad (4.27)$$

Then the following estimates hold uniformly for $|y| \leq \eta$.

1. If $s = \sigma + it \in \mathcal{D}$, $|t| \geq 2$, and $1 - \delta' \leq \sigma \leq 1 + \delta'$, then

$$|F(s, y)| \ll_{\eta, \delta', \varepsilon} |t|^B. \quad (4.28)$$

2. If $T \geq 1$, $s = 1 - \lambda_T + it$, and $|t| \leq 2$, then

$$|F(s, y)| \ll_{\eta, \delta'} \lambda_T^{-2-2\eta} \ll_{\eta, \delta'} (\log(T+2))^{2+2\eta}. \quad (4.29)$$

Proof. For the first estimate, let $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$ and $1 - \delta' \leq \sigma \leq 1 + \delta'$. On $\mathcal{D}$, the chosen branch gives

$$F(s, y) = \zeta(s)^\alpha H(s, y), \quad \zeta(s)^\alpha = \exp(\alpha L(s)).$$

Writing $L(s) = \log |\zeta(s)| + i \arg \zeta(s)$ on this branch,

$$|\zeta(s)^\alpha| = |\zeta(s)|^{\operatorname{Re}(\alpha)} \exp(-\Im(\alpha) \arg \zeta(s)).$$

Since $|y| \leq \eta$, we have $\operatorname{Re}(\alpha) \leq 2 + 2\eta$ and $|\Im(\alpha)| \leq 2\eta$. Hence

$$|F(s, y)| \leq |H(s, y)| (1 + |\zeta(s)|)^{2+2\eta} \exp(2\eta |\arg \zeta(s)|).$$

Here the factor $1 + |\zeta(s)|$ avoids any lower-bound assumption on $|\zeta(s)|$. Proposition 3.4, applied with $\delta = \delta'$, gives $|H(s, y)| \ll_{\eta, \delta'} 1$. Moreover, Theorem 4.3 gives

$$|\zeta(s)| \ll_{\delta', \varepsilon} |t|^{\delta'/2+\varepsilon},$$

and Theorem 4.4 gives

$$|\arg \zeta(s)| \leq C_{\arg} \log |t|.$$

Since $|t| \geq 2$, these estimates imply

$$|F(s, y)| \ll_{\eta, \delta', \varepsilon} |t|^{(2+2\eta)(\delta'/2+\varepsilon)+2\eta C_{\arg}} = |t|^B,$$

which proves (4.28).

We now prove the low-height estimate. Let

$$q(s) := (s-1)\zeta(s), \quad q(1) := 1.$$

Then q is holomorphic at $s = 1$. Put

$$\lambda_0 := \frac{c_0}{2 \log 3}, \quad K := \{s = \sigma + it : 1 - \lambda_0 \leq \sigma \leq 1, |t| \leq 2\}.$$

For $|t| \leq 2$ we have $\log(|t|+2) \leq \log 4 < 2 \log 3$, and hence

$$\lambda_0 = \frac{c_0}{2 \log 3} < \frac{c_0}{\log(|t|+2)}.$$

Thus $K \setminus \{1\}$ lies in the zero-free region of Theorem 4.2. After setting $q(1) = 1$, the function q is holomorphic and non-vanishing on K . Since K is compact, there is an open neighborhood U of K on which q is non-vanishing. Choose a holomorphic branch

$$M(s) = \log q(s)$$

on U , compatible with $M(s) = L(s) + \text{Log}(s-1)$ on $U \cap \mathcal{D}$. Let

$$M_K := \sup_{s \in K} |M(s)| < \infty.$$

For $|y| \leq \eta$,

$$|q(s)^\alpha| = |\exp(\alpha M(s))| \leq \exp(|\alpha| |M(s)|) \leq \exp((2+2\eta)M_K) \quad (s \in K).$$

Define

$$G(s, y) := q(s)^\alpha H(s, y). \quad (4.30)$$

Then, on the chosen branch near the slit,

$$F(s, y) = G(s, y)(s-1)^{-\alpha}. \quad (4.31)$$

Since $K \subset \{s : \text{Re}(s) \geq 1 - \delta'\}$, combining the preceding bound for $q(s)^\alpha$ with Proposition 3.4, again with $\delta = \delta'$, gives

$$|G(s, y)| = |q(s)^\alpha H(s, y)| \ll_{\eta, \delta'} 1 \quad (s \in K, |y| \leq \eta).$$

On the left vertical line $s = 1 - \lambda_T + it$ with $|t| \leq 2$, we use (4.31). For either boundary value of the logarithm along the slit, $|\text{Arg}(s-1)| \leq \pi$, so

$$|(s-1)^{-\alpha}| = |s-1|^{-\text{Re}(\alpha)} \exp(\Im(\alpha) \text{Arg}(s-1)) \ll_\eta |s-1|^{-\text{Re}(\alpha)}.$$

Since $|s-1| \geq \lambda_T$, if $|s-1| \leq 1$ then

$$|s-1|^{-\text{Re}(\alpha)} \leq |s-1|^{-2-2\eta} \leq \lambda_T^{-2-2\eta}.$$

If $|s-1| \geq 1$, then $\text{Re}(\alpha) \geq 2-2\eta > 0$, so $|s-1|^{-\text{Re}(\alpha)} \leq 1 \leq \lambda_T^{-2-2\eta}$. Hence

$$|F(1 - \lambda_T + it, y)| \ll_{\eta, \delta'} \lambda_T^{-2-2\eta}.$$

The definition (4.14) gives

$$\lambda_T^{-2-2\eta} = \left(\frac{2 \log(T+2)}{c_0} \right)^{2+2\eta} \ll_{\eta, \delta'} (\log(T+2))^{2+2\eta},$$

which proves (4.29). □

Since

$$B(\eta, \delta', \varepsilon) = (2+2\eta) \left(\frac{\delta'}{2} + \varepsilon \right) + 2\eta C_{\text{arg}} \longrightarrow \delta' \quad (\eta, \varepsilon \rightarrow 0^+),$$

and $\delta' < c_0/2$, we may choose $\eta > 0$ and $\varepsilon > 0$ sufficiently small so that

$$B < c_0/2. \quad (4.32)$$

This choice is fixed throughout the contour estimates.

Nonlocal contour pieces

Lemma 4.6 (Bound for the horizontal contour pieces). For all sufficiently large x and every $T \geq 2$,

$$|I_{\text{top}}| + |I_{\text{bot}}| \ll_{\eta, \varepsilon} x^c T^{B-1}, \quad (4.33)$$

uniformly in $|y| \leq \eta$.

Proof. On $\mathcal{C}_{\text{top}}$, parameterize $s = \sigma + iT$ with $\sigma_0(T) \leq \sigma \leq c$. Then

$$\begin{aligned} |I_{\text{top}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\text{top}}} F(s, y) \frac{x^s}{s} ds \right| \\ &\leq \frac{1}{2\pi} \int_{\sigma_0(T)}^c \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right| d\sigma \\ &\leq \frac{c - \sigma_0(T)}{2\pi} \sup_{\sigma_0(T) \leq \sigma \leq c} \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right|. \end{aligned}$$

On this segment, $|x^{\sigma+iT}| = x^\sigma \leq x^c$ and $|\sigma + iT| \geq T$. Also $\sigma_0(T) \geq 1 - \delta'$ by (4.15), while $c \leq 1 + \delta'$ for all sufficiently large x . Hence Lemma 4.5 gives

$$\sup_{\sigma_0(T) \leq \sigma \leq c} |F(\sigma + iT, y)| \ll_{\eta, \varepsilon} T^B.$$

Since

$$c - \sigma_0(T) = \frac{1}{\log x} + \lambda_T = \frac{1}{\log x} + \frac{c_0}{2 \log(T+2)} \ll 1,$$

we obtain

$$\begin{aligned} |I_{\text{top}}| &\ll_{\eta, \varepsilon} \frac{x^c}{T} \sup_{\sigma_0(T) \leq \sigma \leq c} |F(\sigma + iT, y)| \\ &\ll_{\eta, \varepsilon} x^c T^{B-1}. \end{aligned}$$

The same computation on $\mathcal{C}_{\text{bot}}$, with $s = \sigma - iT$, gives

$$|I_{\text{bot}}| \ll_{\eta, \varepsilon} x^c T^{B-1}.$$

This proves (4.33). □

Lemma 4.7 (Bound for the left vertical contour piece). For every $T \geq 2$,

$$|I_{\text{left}}| \ll_{\eta, \varepsilon} x^{1-\lambda_T} T^B, \quad (4.34)$$

uniformly in $|y| \leq \eta$.

Proof. The contour $\mathcal{C}_{\text{left}} = \mathcal{C}_{\text{left}}^- \cup \mathcal{C}_{\text{left}}^+$ is estimated by parameterizing the two boundary-value pieces as $s = \sigma_0(T) + it$, with $-T \leq t < 0$ on $\mathcal{C}_{\text{left}}^-$ and $0 < t \leq T$ on $\mathcal{C}_{\text{left}}^+$. Using $\sigma_0(T) = 1 - \lambda_T$, we first write

$$\begin{aligned} |I_{\text{left}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\text{left}}} F(s, y) \frac{x^s}{s} ds \right| \\ &\leq \frac{1}{2\pi} \left(\int_{-T}^{0^-} \left| F(\sigma_0(T) + it, y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T) + it} \right| dt + \int_{0^+}^T \left| F(\sigma_0(T) + it, y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T) + it} \right| dt \right) \\ &= \frac{x^{1-\lambda_T}}{2\pi} \left(\int_{-T}^{0^-} \left| \frac{F(\sigma_0(T) + it, y)}{\sigma_0(T) + it} \right| dt + \int_{0^+}^T \left| \frac{F(\sigma_0(T) + it, y)}{\sigma_0(T) + it} \right| dt \right). \end{aligned}$$

Since $\sigma_0(T) > 1 - \delta' > 1/2$, we have $|\sigma_0(T) + it| \gg 1$ for $|t| \leq 2$ and $|\sigma_0(T) + it| \gg |t|$ for $2 \leq |t| \leq T$. Hence

$$|I_{\text{left}}| \ll x^{1-\lambda_T} \left(\int_{|t| \leq 2} |F(\sigma_0(T) + it, y)| dt + \int_{2 \leq |t| \leq T} \frac{|F(\sigma_0(T) + it, y)|}{|t|} dt \right).$$

On $|t| \leq 2$, Lemma 4.5 gives

$$|F(\sigma_0(T) + it, y)| \ll_{\eta, \delta'} (\log(T+2))^{2+2\eta}.$$

On $2 \leq |t| \leq T$, the point $\sigma_0(T) + it$ lies in $\mathcal{D}$ as a boundary-value limit and satisfies $1 - \delta' \leq \sigma_0(T) \leq 1 + \delta'$, so Lemma 4.5 gives

$$|F(\sigma_0(T) + it, y)| \ll_{\eta, \varepsilon} |t|^B.$$

Therefore

$$\begin{aligned} |I_{\text{left}}| &\ll_{\eta, \delta', \varepsilon} x^{1-\lambda_T} \left((\log(T+2))^{2+2\eta} + \int_{2 \leq |t| \leq T} |t|^{B-1} dt \right) \\ &\ll_{\eta, \delta', \varepsilon} x^{1-\lambda_T} \left((\log(T+2))^{2+2\eta} + 2 \int_2^T t^{B-1} dt \right). \end{aligned}$$

Since $B > 0$,

$$2 \int_2^T t^{B-1} dt = \frac{2}{B} (T^B - 2^B) \ll_B T^B.$$

Also $(\log(T+2))^{2+2\eta} \ll_{\eta, B} T^B$ for $T \geq 2$. Combining these estimates gives

$$|I_{\text{left}}| \ll_{\eta, \delta', \varepsilon} x^{1-\lambda_T} T^B.$$

This proves (4.34). □

Choice of T and the combined error

We choose

$$T = T(x) := \exp(\sqrt{\log x}), \tag{4.35}$$

the standard Selberg–Delange truncation height adapted to the de la Vallée Poussin zero-free region. With this choice, $\log T = \sqrt{\log x}$, and $\lambda_T = c_0/(2\sqrt{\log x} + O(1))$, so $\lambda_T < \delta'$ holds automatically. Moreover, $c = 1 + 1/\log x \leq 1 + \delta'$ for all x sufficiently large, so the abscissa c lies inside the strip used in the growth estimates. We have

$$x^{-\lambda_T} = \exp(-\lambda_T \log x) = \exp\left(-\frac{c_0}{2} \sqrt{\log x} \cdot (1 + o(1))\right). \tag{4.36}$$

Lemma 4.8 (Short interval bound for d_K). For each fixed integer $K \geq 1$, there exists a constant $C_K > 0$ such that, for all sufficiently large X , if

$$C_K X^{1-1/(2K)} \leq H \leq X,$$

then

$$\sum_{X < n \leq X+H} d_K(n) \ll_K H(\log X)^{K-1}.$$

Here $d_K(n)$ is defined by

$$\zeta(s)^K = \sum_{n \geq 1} d_K(n) n^{-s} \quad (\operatorname{Re}(s) > 1),$$

or equivalently as the number of ordered K -tuples of positive integers whose product is n .

Proof. We prove the assertion by induction on K . For $K = 1$, we have $d_1(n) = 1$, so the estimate is immediate.

Assume $K \geq 2$ and that the result is known for $K - 1$. Put

$$D_K(X, H) := \sum_{X < n \leq X+H} d_K(n).$$

This counts ordered K -tuples $(m_1, \dots, m_K)$ satisfying

$$X < m_1 \cdots m_K \leq X + H.$$

Since $H \leq X$, the product is at most $2X$, so at least one factor is at most $(2X)^{1/K}$. By the union bound over the K possible positions of such a factor,

$$D_K(X, H) \leq K \sum_{m \leq (2X)^{1/K}} \sum_{X/m < r \leq (X+H)/m} d_{K-1}(r).$$

Let $Y = X/m$ and $L = H/m$. Choose C_K large enough, in terms of K and C_{K-1} , so that whenever $H \geq C_K X^{1-1/(2K)}$ and $m \leq (2X)^{1/K}$, we have

$$L \geq C_{K-1} Y^{1-1/(2(K-1))}.$$

Also $L \leq Y$, since $H \leq X$. Hence the induction hypothesis applies to the inner sum and gives

$$\sum_{X/m < r \leq (X+H)/m} d_{K-1}(r) \ll_K \frac{H}{m} (\log X)^{K-2}.$$

Therefore

$$D_K(X, H) \ll_K H (\log X)^{K-2} \sum_{m \leq (2X)^{1/K}} \frac{1}{m} \ll_K H (\log X)^{K-1}.$$

□

Lemma 4.9 (Perron truncation error). Let $T = T(x) = \exp(\sqrt{\log x})$ and $c = 1 + 1/\log x$. For $x \geq 2$ with $x \notin \mathbb{Z}$ and $|y| \leq \eta$, there exists a fixed integer $K = K(\eta)$ such that

$$E_{\text{Perron}}(x, T, y) \ll_{\eta} \frac{x}{T} (\log x)^K. \quad (4.37)$$

Proof. Set $M = 1 + \eta$ and choose an integer $K = K(\eta)$ with $K \geq 2M$ and $K \geq 2$. We first show that

$$d(n)M^{\omega(n)} \leq d_K(n) \quad (n \geq 1).$$

For a prime power p^a with $a \geq 1$,

$$d(p^a)M^{\omega(p^a)} = (a+1)M, \quad d_K(p^a) = \binom{a+K-1}{K-1}.$$

The quantity

$$\frac{1}{a+1} \binom{a+K-1}{K-1}$$

is increasing in a for $K \geq 2$, and at $a = 1$ it equals $K/2$. Thus $(a+1)M \leq d_K(p^a)$ for every $a \geq 1$. Extending multiplicatively gives the claimed coefficient domination.

Using (4.6), we obtain

$$E_{\text{Perron}}(x, T, y) \ll_{\eta} x^c \sum_{n \geq 1} \frac{d_K(n)}{n^c} \frac{1}{1 + T \left| \log \frac{x}{n} \right|}.$$

We split the sum into the ranges $n \leq x/2$, $x/2 < n < 2x$, and $n \geq 2x$.

If $n \leq x/2$ or $n \geq 2x$, then $|\log(x/n)| \geq \log 2$, and hence the weight is $\ll 1/T$. The contribution of these two far ranges is therefore

$$\ll \frac{x^c}{T} \sum_{n \geq 1} \frac{d_K(n)}{n^c} = \frac{x^c}{T} \zeta(c)^K.$$

Since $c = 1 + 1/\log x$, we have $x^c \asymp x$ and $\zeta(c) \ll \log x$, so the far contribution is

$$\ll_K \frac{x}{T} (\log x)^K.$$

It remains to estimate the central range $x/2 < n < 2x$. There $x^c n^{-c} = (x/n)^c \ll 1$, so it is enough to bound

$$\sum_{x/2 < n < 2x} d_K(n) \frac{1}{1 + T \left| \log \frac{x}{n} \right|}.$$

Put $H = x/T$. Cover the central range by the right intervals

$$I_r^+ = (x + rH, x + (r+1)H]$$

and the left intervals

$$I_r^- = [x - (r+1)H, x - rH),$$

with $r = 0, 1, 2, \dots$ and only $O(T)$ intervals needed. For $n \in I_r^{\pm}$,

$$\frac{1}{1 + T \left| \log \frac{x}{n} \right|} \ll \frac{1}{1 + r}.$$

For $r = 0$ this is trivial. For $r \geq 1$, since $x/2 < n < 2x$, we have

$$\left| \log \frac{x}{n} \right| \asymp \frac{|n - x|}{x} \asymp \frac{rH}{x} = \frac{r}{T}.$$

Because $T = \exp(\sqrt{\log x})$, the length $H = x \exp(-\sqrt{\log x})$ is larger than $C_K x^{1-1/(2K)}$ for all sufficiently large x . Each interval lies in $x/2 < n < 2x$, so Lemma 4.8 applies with a starting point comparable to x , and gives

$$\sum_{n \in I_r^{\pm}} d_K(n) \ll_K H (\log x)^{K-1}.$$

Therefore the central contribution is

$$\ll_K H (\log x)^{K-1} \sum_{r \ll T} \frac{1}{1 + r} \ll_K \frac{x}{T} (\log x)^{K-1} \log T.$$

Since $\log T = \sqrt{\log x} \leq \log x$ for sufficiently large x , this is

$$\ll_K \frac{x}{T} (\log x)^K.$$

Combining the far and central estimates proves (4.37). □

Proposition 4.10. With $T = T(x) = \exp(\sqrt{\log x})$, there exists a constant $c_1 > 0$ (depending on η, ε but not on x) such that

$$E_{\text{Perron}}(x, T, y) + |I_{\text{top}}| + |I_{\text{bot}}| + |I_{\text{left}}| \ll_{\eta, \varepsilon} x \exp(-c_1 \sqrt{\log x}), \quad (4.38)$$

uniformly in $|y| \leq \eta$, for all sufficiently large $x \notin \mathbb{Z}$.

Proof. We bound each term separately and take the worst.

Perron error. By Lemma 4.9 with $T = \exp(\sqrt{\log x})$,

$$E_{\text{Perron}} \ll_{\eta} x \exp(-\sqrt{\log x})(\log x)^K.$$

Since $(\log x)^K = \exp(K \log \log x)$ and, for sufficiently large x ,

$$K \log \log x \leq \frac{1}{2} \sqrt{\log x},$$

we have

$$E_{\text{Perron}} \ll_{\eta} x \exp\left(-\frac{1}{2} \sqrt{\log x}\right).$$

Horizontal pieces. With $c = 1 + 1/\log x$ so that $x^c = e \cdot x$, by Lemma 4.6,

$$|I_{\text{top}}| + |I_{\text{bot}}| \ll_{\eta, \varepsilon} x \cdot T^{B-1} = x \cdot \exp((B-1)\sqrt{\log x}).$$

Since $B < 1$, this is $\ll x \exp(-(1-B)\sqrt{\log x})$.

Vertical piece. By Lemma 4.7 and (4.36),

$$|I_{\text{left}}| \ll_{\eta, \varepsilon} x \cdot x^{-\lambda_T} \cdot T^B = x \cdot \exp\left(-\frac{c_0}{2} \sqrt{\log x} + B \sqrt{\log x}\right)(1 + o(1)).$$

By the choice (4.13) of δ' together with (4.32), we have $B < c_0/2$. Hence $|I_{\text{left}}| \ll x \exp(-(c_0/2 - B)\sqrt{\log x})$.

Set

$$c_1 := \frac{1}{2} \min\left(\frac{1}{2}, 1 - B, \frac{c_0}{2} - B\right) > 0.$$

For all sufficiently large x , the preceding three estimates are bounded by the right-hand side of (4.38). $\square$

The right-hand side of (4.38) is exponentially smaller than the local contribution on the $A(x, y)$ level. The local Hankel contribution I_H , analyzed in Section 5, contains the main term from which the asymptotic formula for each fixed $T_j(x)$ will later be extracted.

5 Local Hankel analysis

The purpose of this section is to analyze the local Hankel contribution I_H around the singular point $s = 1$. Recall that

$$F(s, y) = \zeta(s)^{2+2y} H(s, y).$$

The singular part is carried by the power of $\zeta(s)$; the remaining factor should be made regular at $s = 1$ before the local Hankel integral is evaluated.

5.1 The local regular factor

We keep the abbreviation

$$\alpha = 2 + 2y.$$

Define

$$q(s) := (s - 1)\zeta(s), \quad q(1) := 1. \quad (5.1)$$

The pole of $\zeta(s)$ at $s = 1$ is simple with residue 1, so q is holomorphic near $s = 1$ after filling in the removable singularity. After shrinking the neighborhood of $s = 1$ if necessary, we may assume that $q(s) \neq 0$ there. We then choose a single branch of $\log q(s)$ in this neighborhood, normalized by

$$\log q(1) = 0.$$

For s in this neighborhood and y near 0, define

$$G(s, y) := q(s)^{2+2y} H(s, y), \quad q(s)^{2+2y} := \exp((2 + 2y) \log q(s)). \quad (5.2)$$

On the corresponding punctured slit neighborhood of $s = 1$, the branch choices from Section 4 give

$$F(s, y) = G(s, y)(s - 1)^{-2-2y}. \quad (5.3)$$

Here the branch of $(s - 1)^{-2-2y}$ is the local Hankel branch induced by the contour and branch conventions already fixed in Section 4.3.

Lemma 5.1 (Analyticity and boundedness of the local factor). Fix $0 < \eta < 1/2$ and $0 < \delta < 1/2$. There exist $r \in (0, \delta)$ and $C_{\eta, \delta} > 0$ such that $G(s, y)$ is holomorphic in (s, y) for $|s - 1| < r$ and $|y| < \eta$, and

$$|G(s, y)| \leq C_{\eta, \delta}$$

uniformly for $|s - 1| \leq r$ and $|y| \leq \eta$. Moreover, on the punctured slit neighborhood

$$0 < |s - 1| < r, \quad s \notin \mathcal{S},$$

we have the local factorization

$$F(s, y) = G(s, y)(s - 1)^{-2-2y}.$$

Proof. Apply Proposition 3.4 with this fixed δ . It gives that the function $H(s, y)$ is holomorphic in (s, y) on the interior of $\mathcal{R}_{\delta, \eta}$ and uniformly bounded on $\mathcal{R}_{\delta, \eta}$. Choose $r \in (0, \delta)$. Then $|s - 1| \leq r$ implies $\operatorname{Re}(s) \geq 1 - r > 1 - \delta$, so

$$H(s, y) \ll_{\eta, \delta} 1 \quad (|s - 1| \leq r, |y| \leq \eta).$$

The function $q(s) = (s - 1)\zeta(s)$ is holomorphic at $s = 1$ after setting $q(1) = 1$. Since $q(1) = 1 \neq 0$, we shrink r , preserving $r \in (0, \delta)$, so that q has no zeros on $|s - 1| \leq r$. Then a holomorphic branch of $\log q(s)$ exists on $|s - 1| < r$, normalized by $\log q(1) = 0$. Since this branch is holomorphic on a neighborhood of the closed disk $|s - 1| \leq r$, it is bounded there. Therefore

$$q(s)^{2+2y} = \exp((2 + 2y) \log q(s))$$

is uniformly bounded for $|s - 1| \leq r$ and $|y| \leq \eta$. Combining this local bound with the bound for $H(s, y)$ gives $|G(s, y)| \leq C_{\eta, \delta}$.

Finally, on the punctured slit neighborhood,

$$\zeta(s)^{2+2y} = q(s)^{2+2y}(s - 1)^{-2-2y},$$

with the branch of $(s - 1)^{-2-2y}$ inherited from the local Hankel contour. Combining this identity with $F(s, y) = \zeta(s)^{2+2y} H(s, y)$ proves the asserted factorization. $\square$

Lemma 5.2 (First-order approximation of the local factor). Fix $0 < \eta < 1/2$ and $0 < \delta < 1/2$, and let $r \in (0, \delta)$ be chosen as in Lemma 5.1. Then

$$G(1, y) = H(1, y)$$

uniformly for $|y| \leq \eta$, and

$$G(1, 0) = 1.$$

Moreover,

$$G(s, y) = G(1, y) + O_{\eta, \delta}(|s - 1|)$$

uniformly for $|s - 1| \leq r/2$ and $|y| \leq \eta$.

Proof. Since $q(1) = 1$ and the branch is normalized by $\log q(1) = 0$, we have

$$G(1, y) = q(1)^{2+2y} H(1, y) = H(1, y).$$

By Lemma 3.5, $H(1, 0) = 1$. Therefore $G(1, 0) = H(1, 0) = 1$.

For the first-order estimate, use Lemma 5.1. Since G is holomorphic in s on $|s - 1| < r$ and uniformly bounded for $|s - 1| \leq r$, $|y| \leq \eta$, Cauchy's estimate in the s -variable gives

$$\frac{\partial G}{\partial s}(s, y) \ll_{\eta, \delta} 1$$

uniformly for $|s - 1| \leq r/2$ and $|y| \leq \eta$. Therefore

$$G(s, y) - G(1, y) = \int_1^s \frac{\partial G}{\partial s}(w, y) dw = O_{\eta, \delta}(|s - 1|),$$

uniformly in the stated range. □

5.2 Evaluation of the local Hankel contribution

Proposition 5.3 (Local Hankel contribution). With η fixed as in Section 4, put

$$T = \exp(\sqrt{\log x}), \quad \rho := \frac{1}{\log x}.$$

For all sufficiently large x , uniformly for $|y| \leq \eta$, we have

$$I_H = x H(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_{\eta}(x(\log x)^{2\eta}).$$

Proof. Put $\alpha = 2 + 2y$. We apply Lemmas 5.1 and 5.2 with the fixed strip depth δ' from Section 4.3; since δ' is fixed, the constants below depend only on η . Let r be the local radius supplied by these lemmas. With $T = \exp(\sqrt{\log x})$, we have

$$\lambda_T = \frac{c_0}{2 \log(T+2)} \asymp \frac{1}{\sqrt{\log x}}.$$

Thus $\rho = 1/\log x < \lambda_T$ and $\rho < r/2$ for all sufficiently large x .

By the definition of I_H in (4.26), together with the local factorization recorded in Lemma 5.1,

$$\begin{aligned} I_H &:= \frac{1}{2\pi i} \int_{\mathcal{H}_{\text{loc}}} F(s, y) \frac{x^s}{s} ds \\ &= \frac{1}{2\pi i} \int_{\mathcal{H}_{\text{loc}}} G(s, y) (s-1)^{-\alpha} \frac{x^s}{s} ds \\ &= \frac{x}{2\pi i} \int_{\mathcal{H}_z} \frac{G(1+z, y)}{1+z} z^{-\alpha} e^{z \log x} dz, \end{aligned}$$

where the last equality follows from the change of variables $z = s - 1$. Here $\mathcal{H}_z$ is the image of $\mathcal{H}_{\text{loc}}$ under this change of variables. Thus $\mathcal{H}_z$ is the truncated Hankel contour whose arms run along the negative real axis from $-\lambda_T$ to $-\rho$, with the circle $|z| = \rho$ oriented counterclockwise through the right side of the origin.

For z on $\mathcal{H}_z$, Lemma 5.2 gives

$$G(1+z, y) = G(1, y) + O_\eta(|z|).$$

Also, since $|z| \leq \lambda_T < 1/2$ for large x ,

$$\frac{1}{1+z} = 1 + O(|z|).$$

Using the uniform boundedness of $G(1, y)$, we obtain

$$\frac{G(1+z, y)}{1+z} = G(1, y) + E(z, y), \quad E(z, y) = O_\eta(|z|), \quad (5.4)$$

uniformly for $|y| \leq \eta$ and $z \in \mathcal{H}_z$.

Accordingly, write

$$I_H = M_H + R_H,$$

where

$$M_H := \frac{xG(1, y)}{2\pi i} \int_{\mathcal{H}_z} z^{-\alpha} e^{z \log x} dz$$

and

$$R_H := \frac{x}{2\pi i} \int_{\mathcal{H}_z} E(z, y) z^{-\alpha} e^{z \log x} dz.$$

We first evaluate M_H . Put

$$U := \lambda_T \log x.$$

Under the change of variables $w = z \log x$, the contour $\mathcal{H}_z$ is mapped to a truncated Hankel contour $\mathcal{H}_U$ in the w -plane, whose arms run from $-U$ to -1 and whose circular part is $|w| = 1$. Then

$$M_H = xG(1, y)(\log x)^{\alpha-1} \frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha} dw.$$

Let $\mathcal{H}_\infty$ be the corresponding infinite Hankel contour. By the standard Hankel representation of $1/\Gamma(\alpha)$ [1],

$$\frac{1}{2\pi i} \int_{\mathcal{H}_\infty} e^w w^{-\alpha} dw = \frac{1}{\Gamma(\alpha)}.$$

The difference between $\mathcal{H}_\infty$ and $\mathcal{H}_U$ consists of the upper and lower tails over $(-\infty, -U]$. On these tails write $w = -u \pm i0$, with $u \geq U$. Then $|e^w| = e^{-u}$. Also, since $|y| \leq \eta$ and $\alpha = 2 + 2y$, the branch factors $e^{\pm i\pi\alpha}$ are bounded by a constant depending only on η , and hence

$$|w^{-\alpha}| \ll_\eta u^{-\operatorname{Re}(\alpha)}.$$

After taking absolute values, the two tails and the factor $1/(2\pi i)$ are absorbed into the O_η -constant, giving

$$\left| \frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha} dw - \frac{1}{\Gamma(\alpha)} \right| \ll_\eta \int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)} du.$$

Since

$$\operatorname{Re}(\alpha) = 2 + 2\operatorname{Re}(y) \geq 2 - 2\eta > 1$$

and $U \geq 1$ for sufficiently large x , we have

$$\int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)} du \leq \int_U^\infty e^{-u} du = e^{-U}.$$

Therefore

$$\frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha} dw = \frac{1}{\Gamma(\alpha)} + O_\eta(e^{-U}).$$

It follows that

$$M_H = xG(1, y) \frac{(\log x)^{\alpha-1}}{\Gamma(\alpha)} + O_\eta \left(x(\log x)^{\operatorname{Re}(\alpha)-1} e^{-U} \right).$$

Since $U = \lambda_T \log x \asymp \sqrt{\log x}$, there is a constant $c > 0$ such that $e^{-U} \ll e^{-c\sqrt{\log x}}$. Also $\operatorname{Re}(\alpha) - 1 \leq 1 + 2\eta$. Hence the tail contribution is

$$\ll_\eta x(\log x)^{1+2\eta} e^{-c\sqrt{\log x}} = O_\eta \left(x(\log x)^{2\eta} \right),$$

because $(\log x)e^{-c\sqrt{\log x}} = O(1)$.

It remains to estimate R_H . From (5.4),

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z| |z|^{-\alpha} |e^{\operatorname{Re}(z) \log x}| |dz|.$$

On the Hankel contour, $|\operatorname{Arg} z| \leq \pi$ and $|\Im \alpha| \leq 2\eta$, so

$$|z^{-\alpha}| \ll_\eta |z|^{-\operatorname{Re}(\alpha)}.$$

Thus

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z|^{1-\operatorname{Re}(\alpha)} e^{\operatorname{Re}(z) \log x} |dz|.$$

Changing variables again by $w = z \log x$ gives

$$|R_H| \ll_\eta x(\log x)^{\operatorname{Re}(\alpha)-2} \int_{\mathcal{H}_U} |w|^{1-\operatorname{Re}(\alpha)} e^{\operatorname{Re} w} |dw|.$$

The last integral is $O_\eta(1)$: this is immediate on $|w| = 1$, and on the arms it is bounded by

$$\int_1^U u^{1-\operatorname{Re}(\alpha)} e^{-u} du \leq \int_1^\infty u^{1-\operatorname{Re}(\alpha)} e^{-u} du \ll_\eta 1.$$

Consequently,

$$R_H = O_\eta \left(x(\log x)^{\operatorname{Re}(\alpha)-2} \right).$$

Since $\alpha = 2 + 2y$, we have

$$\operatorname{Re}(\alpha) - 2 = 2 \operatorname{Re}(y) \leq 2\eta,$$

and hence

$$R_H = O_\eta(x(\log x)^{2\eta}).$$

Combining the estimates for M_H and R_H , and using Lemma 5.2 to replace $G(1, y)$ by $H(1, y)$, we obtain

$$I_H = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}),$$

uniformly for $|y| \leq \eta$. □

Proposition 5.4 (Generating-function asymptotic). With the parameters chosen as in Sections 4 and 5, and with the fixed contour parameters suppressed in the implied constant, uniformly for $|y| \leq \eta$, as $x \rightarrow \infty$,

$$A(x, y) = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}).$$

Proof. For non-integral x , the contour decomposition (4.24) gives

$$A(x, y) = I_H + I_{\text{top}} + I_{\text{bot}} + I_{\text{left}} + E_{\text{Perron}}(x, T, y).$$

By Proposition 4.10, for some $c_1 > 0$,

$$E_{\text{Perron}}(x, T, y) + |I_{\text{top}}| + |I_{\text{bot}}| + |I_{\text{left}}| \ll_\eta x \exp(-c_1 \sqrt{\log x}),$$

uniformly for $|y| \leq \eta$. Proposition 5.3 gives

$$I_H = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}).$$

The exponentially small nonlocal and Perron error is absorbed into $O_\eta(x(\log x)^{2\eta})$. Therefore the asserted asymptotic holds for non-integral x .

For integral x , the replacement by $x + 1/2$ described in Section 4.1 gives the same estimate, since $A(x, y) = A(x + 1/2, y)$ and replacing $x + 1/2$ by x changes the displayed main term by an amount absorbed into the same error term. □

6 Extraction of $T_j(x)$ and proof of the main theorem

Proof of Theorem 1.3. By Proposition 5.4, uniformly for $|y| \leq \eta$,

$$A(x, y) = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}).$$

Define

$$K(y) := \frac{H(1, y)}{\Gamma(2+2y)}.$$

Then K is holomorphic in a neighborhood of 0, and

$$A(x, y) = x \log x K(y) e^{2y \log \log x} + O_\eta(x(\log x)^{2\eta}).$$

The generating relation

$$A(x, y) = \sum_{j \geq 0} T_j(x) y^j$$

gives the coefficient identity below. Here $[y^j]$ denotes the coefficient of y^j in the power series expansion:

$$T_j(x) = [y^j] A(x, y).$$

Write

$$K(y) = \sum_{r \geq 0} k_r y^r, \quad k_r = \frac{K^{(r)}(0)}{r!},$$

and

$$e^{2y \log \log x} = \sum_{m \geq 0} \frac{(2 \log \log x)^m}{m!} y^m.$$

For fixed j ,

$$[y^j] \left(K(y) e^{2y \log \log x} \right) = \sum_{r=0}^j k_r \frac{(2 \log \log x)^{j-r}}{(j-r)!}.$$

Therefore

$$[y^j] \left(K(y) e^{2y \log \log x} \right) = k_0 \frac{(2 \log \log x)^j}{j!} + k_1 \frac{(2 \log \log x)^{j-1}}{(j-1)!} + O_j((\log \log x)^{j-2}).$$

It remains to compute k_0 and k_1 . By Lemma 3.5, $H(1, 0) = 1$. Also $\Gamma(2) = 1$. Hence

$$k_0 = K(0) = 1.$$

Next,

$$k_1 = K'(0) = \partial_y H(1, 0) + H(1, 0) \frac{d}{dy} \frac{1}{\Gamma(2 + 2y)} \Big|_{y=0}.$$

Here $\psi = \Gamma'/\Gamma$ denotes the digamma function. Since

$$\frac{d}{dy} \frac{1}{\Gamma(2 + 2y)} \Big|_{y=0} = -2\psi(2) = -2(1 - \gamma),$$

we get

$$k_1 = \partial_y H(1, 0) - 2(1 - \gamma).$$

We also need to extract coefficients from the error term. Choose a fixed radius ρ_y with $0 < \rho_y < \eta$. Since the error in Proposition 5.4 is uniform for $|y| \leq \eta$, Cauchy's coefficient estimate on $|y| = \rho_y$ gives, for each fixed j ,

$$[y^j] O_\eta(x(\log x)^{2\eta}) = O_{\eta, j, \rho_y}(x(\log x)^{2\eta}).$$

Because $0 < \eta < 1/2$, for fixed $j \geq 2$,

$$x(\log x)^{2\eta} = O_j(x \log x (\log \log x)^{j-2}).$$

Thus this error is absorbed into the final error term.

Combining the preceding estimates gives, for each fixed $j \geq 2$,

$$\begin{aligned} T_j(x) &= \frac{2^j}{j!} x \log x (\log \log x)^j \\ &\quad + \frac{2^{j-1}}{(j-1)!} (\partial_y H(1, 0) - 2(1 - \gamma)) x \log x (\log \log x)^{j-1} \\ &\quad + O_j(x \log x (\log \log x)^{j-2}). \end{aligned}$$

In particular,

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log \log x)^j.$$

□

7 Toward $D(x)$: outlook

The identity (1.6) expresses $D(x)$ exactly in terms of the prime-power contribution and the alternating finite sum of the quantities $T_j(x)$. Theorem 1.3, however, is a fixed- j asymptotic statement. Since the range of relevant j grows with x , these fixed- j asymptotics cannot be inserted term by term into (1.6) to obtain an asymptotic formula for $D(x)$.

Equivalently, if

$$A(x, y) = \sum_{j \geq 0} T_j(x) y^j,$$

then the alternating part of (1.6) is determined by values of $A(x, y)$ and its y -derivative at $y = -1$. The analytic work in this paper is carried out for y in a small neighborhood of 0. Extending the method toward $D(x)$ would therefore require estimates uniform in j , or a direct analysis of the generating function near $y = -1$.

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